

SOURAJIT MAJUMDER

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EDUCATION

Narula Institute of Technology <i>B.Tech in Computer Science Engineering (AI & ML)</i> CGPA: 8.51/10.0 <i>Relevant Coursework:</i> Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Database Management Systems	2023 – 2027
St. Augustine's Day School, Barrackpore (ISC) <i>Higher Secondary (Grade 12):</i> 94%	2023
St. Augustine's Day School, Barrackpore (ISC) <i>Secondary (Grade 10):</i> 91%	2021

TECHNICAL SKILLS

Languages: Java, Python, JavaScript
AI / ML & Computer Vision: scikit-learn, LangChain, LangGraph, RAG, ChromaDB
Web & Backend: FastAPI, Node.js, Express.js, REST API, MongoDB, PostgreSQL
Tools & Platforms: Git, GitHub, Docker
Core CS: Data Structures & Algorithms, OOP, System Design Fundamentals

PROJECTS

Glaucoma Detection System — Deep Learning, Computer Vision github.com/Sourajit-M/glaucoma-detection-project Live Demo - Built a retinal disease detection project using AI that got 94.5% AUC and 87.1% accuracy on eye scan images. It performed 15% better than older machine learning methods like SVM and Random Forest.. - Used ResNet-18 for classifying eye images and U-Net for detecting optic disc and cup areas. Also added Grad-CAM to show which parts of the image the AI focused on. - Created the complete machine learning pipeline including image preprocessing, data augmentation, cross-validation, and saving the model for real-world use. Technologies Used: Python, PyTorch, ResNet-18, U-Net, Grad-CAM, scikit-learn, OpenCV, Vercel	2026
YouTube RAG Engine github.com/Sourajit-M/youtube-semantic-search Live Demo - Built a smart search and question-answering system that can understand YouTube transcripts and answer questions with proper source references. - Used a Hybrid Retrieval method that combines keyword search and AI-based semantic search to improve accuracy and reduce wrong search results by about 40%. - Created an automatic system to collect YouTube videos and transcripts, divide them into smaller parts, and save them for faster searching. Technologies Used: Python, FastAPI, ChromaDB, LiteLLM, SQLite, React	2025
Local AI Inference Platform — (FastAPI, React, Ollama) github.com/Sourajit-M/local-ai-inference-platform Live Demo - Built a local AI inference platform supporting offline execution of Small Language Models with JWT authentication, persistent chat sessions, structured generation, and experiment tracking. - Designed a benchmarking and evaluation framework measuring TTFT, latency, throughput, and warm/cold start performance across Qwen 2.5 3B, Llama 3.2 3B, and Phi-3 Mini, visualized through a React analytics dashboard. Technologies Used: Python, FastAPI, React, TypeScript, Ollama, SQLite, Pydantic, JWT, Docker	2026

ACHIEVEMENTS & EXTRACURRICULARS

- LeetCode:** Solved more than **450 coding problems** on LeetCode, mainly on topics like dynamic programming, graphs, trees, and binary search to improve problem-solving skills.
- Certification:** Completed a Full Stack Web Development course from 100xDevs, learning React, Node.js, PostgreSQL, and basic system design through projects.
- Participated in Smart India Internal Hackathon (SIH) 2025 and worked with a team to build an AI-based project within a limited time.
- Took part in coding contests like the Kritanj Coding Competition and regularly practices competitive programming on Codeforces.